Finding a Bus Without Trusting Its GPS

A second way to locate a bus, taken from a 2024 network security paper

> The paper is: Paul, Dhal and Mary, *Relaying Vehicle Selection Protocol for a Road Intersection*, IEEE/ACM Transactions on Networking, volume 32, number 3, June 2024, pages 2438 to 2446. The paper is about cars at a traffic junction, not about buses. The idea behind its solution is what carries over.

1. What was wrong with the earlier idea

The earlier idea was this: take the bus's GPS point, check whether it sits on the route, and reject it if it is far off. That idea has three problems.

Problem one: the route map is public. Delhi's route shapes are open data. Anyone can download them. So a bus that wants to lie can put its fake position exactly on the line, and the check will pass. A check that a clever liar passes for free is not really a check.

Problem two: it does not find the bus. It only says yes or no to a position the bus already gave. If the bus stops sending, nothing comes out at all. The question was how to find the bus, not how to argue with it.

Problem three: it guessed forward. The earlier answer also said, in effect, that the bus was here at 9:05 and this stretch usually takes four minutes, so at 9:07 it should be halfway. That only works if the bus keeps to the same route and the same speed. It does not. Two or three routes can serve the same place. The bus may take a different one today. A road may be closed for repairs. Traffic changes hour by hour. So that whole approach is wrong and should be dropped.

> All three problems come from the same mistake: the method was trying to predict where the bus should be, instead of measuring where it is.

2. What the paper actually uses

This is the part that was missed, and it is the part that matters.

The paper uses **no route. No timetable. No speed. No history of any kind.**

It uses exactly two things:

1. Four fixed points whose positions are already known. In the paper these are satellites.
2. The measured distance from the car to each of those four points.

Then it finds the one place on the ground where all four distances fit at the same moment. That place is the car.

The paper never asks what road the car is on. It never asks how fast the car is going. A car taking a diversion, a car stuck in traffic, a car that chose a different route today: all of these are the same to this method, because it measures instead of predicting.

> The car is never asked "where are you?". The system works out where the car must be, using a clock the car does not own.

3. The bus version: cell towers as the fixed points

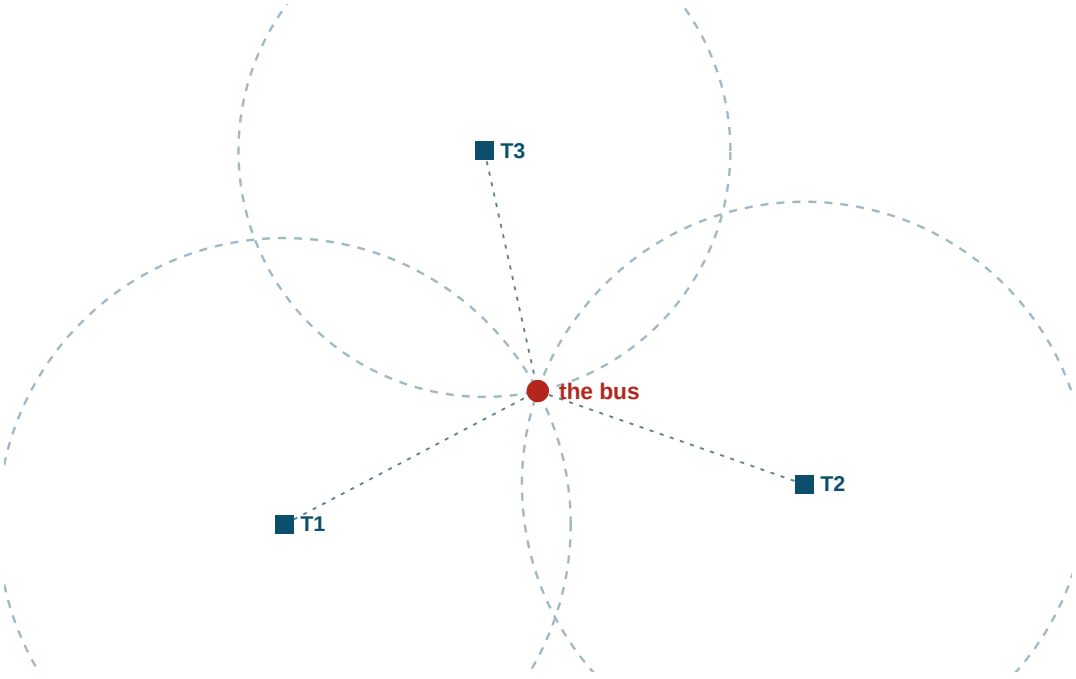
A bus already carries a SIM. The tower it is connected to has to know how far away that phone is. This is not optional. In a mobile network every phone must send its signal so that it arrives at the tower at the same instant as everybody else's. To arrange that, the tower measures how long the phone's signal takes to reach it, and tells the phone to send a little earlier or a little later.

That measured time **is a distance**. The tower works it out. The bus does not report it, and the bus has no way to change it.

So:

1. One tower gives a circle. The bus is somewhere on that circle.
2. Two towers give two circles, which cross at two points.
3. Three towers give three circles, which cross at one point.

That point is the bus. Measured, not reported.



Three towers, three measured distances, one point where all three meet. The bus never reports anything.

> The line that matters: **the tower holds the clock, not the bus.** That is exactly what the satellite does in the paper, and it is the reason the bus cannot lie about it.

4. Why this answers the objections

Objection	Why it stops mattering
Two or three routes serve the same place	There is no route anywhere in the calculation. Geometry gives a point, not a road, so there is no route to choose and no route to choose wrongly.
The same bus may take a different route today	No route was assumed, so nothing can be violated. The method does not care which road the bus is on.
A road is closed or under repair	The method is not predicting a path. It measures where the bus is right now, wherever that turns out to be.
Bus speed is not constant	Speed is never used. Not measured, not assumed, not averaged.
The bus is late, or early, or off schedule	The timetable is not involved either. Nothing is compared against a schedule.

The short version: the earlier method was **guessing forward**. This one is **measuring now**.

5. How to build and test it

Real tower measurements cannot be obtained from a telecom operator. So the method is built and tested the way this project already handles crowdsourcing: as a designed method, checked honestly against real recorded data.

1. **Get real tower locations.** OpenCelliD is free and public, and holds real cell tower positions for Delhi.
2. **Take the recorded archive.** For every bus position already recorded, work out what the distance to the three nearest towers would be.
3. **Add honest noise.** Tower timing is coarse, roughly 78 metres per step. Add that error. Do not hide it.
4. **Run the paper's maths.** Draw the three circles and find where they cross.
5. **Report the error.** Compare the worked-out position with the GPS position, in metres. Give the middle value and the worst case, not one flattering number.
6. **Then attack it.** Move a bus 2 km on the map and show that the tower distances no longer fit, so the lie is caught. Do the same for a slow fake drift.

Step five is worth stressing. This method produces the one number the paper itself never gives: how far off the position actually is. That is a result, not a demonstration.

6. If tower data is not allowed

A weaker version, still free of routes and speeds.

Use only the tower the bus is attached to. That gives one circle, and the bus is somewhere inside it. Now do **not** pick a route. Lay that circle over the whole Delhi road map and keep every road that falls inside it.

The result is a small set of possible places. No route is chosen, no speed is assumed. It answers the multiple-route objection directly: all routes are kept and none is committed to.

7. What this cannot do

- It cannot give metre accuracy. Tower timing is coarse. Expect tens of metres, or a few hundred, depending on how dense the towers are in that part of the city.
- It needs tower measurements that a student cannot obtain. Everything above is therefore a simulation against real data, not a live deployment. Say that plainly rather than letting it be discovered.
- It does not replace GPS. It is a second, independent source, and that is the whole point: when the GPS is wrong or missing, this still says something useful.

8. In one paragraph

> The earlier idea guessed forward from a known point, and assumed the bus keeps its route and its speed. Both assumptions are false. The paper does not guess forward. It takes fixed points with known positions, measures the distance to each, and finds the one point where all the distances fit. There is no route and no speed in it, so a diversion or a jam changes nothing. For a bus, the fixed points are cell towers: the tower measures how long the bus's signal takes to arrive, which is a distance, and the tower does that measuring, not the bus. Three towers give three circles that cross at one point, and that point is the bus. Since real tower data cannot be obtained, the method is built and tested against the recorded archive using free public tower locations and honest timing noise, and the error is reported in metres. That also supplies the result the paper itself never reports.